

Multiple Choice (10 marks)

1. In force-current analogy, electrical analogous quantity for displacement (x) is
 - (a) inductance.
 - (b) resistance.
 - (c) impedance.
 - (d) voltage.
 - (e) capacitance.
2. A 7-hp motor takes 6.3 kilowatts at full load. What is its efficiency ?
 - (a) 75%
 - (b) 86%
 - (c) 95%
 - (d) 60%
 - (e) 80%
3. Find the inverse Laplace transform $L^{-1}[1/\{s(s^2 + 4)\}]$, using the convolution.
 - (a) $(1/2)(e^{2t} - 1)$
 - (b) $(1/2)(1 + \cos 2t)$
 - (c) $(1/4)(1 - \cos 2t)$
 - (d) $(1/4)(\sin(2t) - \cos(2t))$
 - (e) $(1/2)\sin(2t)$
4. The current through an inductor with inductance $L = 10^{-3}\text{henry}$ is given as $i_L(t) = 0.1\sin 10^6 t$. Find the voltage $V_L(t)$ across this inductor.
 - (a) $0.1 \cos 10^6 t$
 - (b) $100 \cos 10^6 t$
 - (c) $1 \cos 10^6 t$
 - (d) $10^{-3}\cos 10^6 t$
 - (e) $0.01 \sin 10^6 t$
5. Find the inverse Laplace transforms (a) $L^{-1}\{2s/(s^2 + 1)^2\}$, (b) $L^{-1}\{1/s\}$
 6. $t \sin t$, $(1/\sqrt{\pi}) \text{erfc}(1/\sqrt{t})$
 7. $\cos t$, $(1/\sqrt{\pi}) \cdot t e^{-t^2}$

8. $e^t \sin t, (1 / 2\sqrt{\pi}) \cdot t$
9. $t \cos t, (1 / \sqrt{\pi})$ ullet $(1 /)$
10. $t^2 \sin t, (1 / 2\sqrt{\pi}) \cdot (1 /)$

True / False (10 marks)

Answer True or False

6. True or False: Silicon and Germanium are monovalent elements, each having one valence electron available for bonding.
7. True or False: The first digit from right to left in the binary number system is referred to as the Most Significant Bit (MSB).
8. True or False: The cutoff frequency of the first higher-order mode for a 50-ohm coaxial line with a 2 mm inner conductor radius is 14.5 GHz.
9. True or False: If the holding current of a thyristor is 2 mA, the latching current should be 0.002 A.
10. True or False: The normal voltage used in directional heating is 1.5 KV.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Given the joint density function $f(x, y, z) = e^{-(x+y+z)}$ for $0 < x, y, z < \infty$, derive the cumulative distribution function $F(x, y, z)$.
11. Discuss the role of multiplexers in the ALU and control section of a CPU, analyzing their significance in data processing and control flow within computer architecture.

End of Paper